

DNS Reconnaissance Deep

Why attackers love DNS

DNS is public, distributed, and usually unfiltered for inbound resolution. It reveals the entire architecture if misconfigured.

The record types that matter

Record	What it leaks
A / AAAA	Host IPs
NS	Nameservers (attack surface + zone controls)
MX	Mail servers (spoofing candidates)
TXT	SPF, DMARC (email security posture), random secrets people paste
SOA	Zone transfer control + serial
CNAME	Aliases; sometimes reveal internal names
PTR	Reverse hints for internal hostnames

Zone walking & transfers (rarely allowed by hosts)

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# Manual
dig axfr example.com @ns1.example.com # usually REFUSED (good)
# Subdomain brute with a good wordlist
dnsrecon -d example.com -t brt -D /usr/share/wordlists/dnsrecon/subdomains-top1million.txt
# Subdomain via HTTP fuzzing
ffuf -w subdomains.txt -u http://FUZZ.example.com
```

IPv6 & PTR tricks

IPv6 ranges are huge – easy to hide hosts

PTR records sometimes expose internal naming conventions

Enumerating internal DNS (if you reach it) = architecture blueprint

DNS and exfiltration

DNS tunneling: encode data in DNS queries to a domain you control (e.g., iodine, dnscat2) – data leaves even where HTTP is blocked

DNS exfil detection: blue teams watch odd subdomain patterns & high query volumes

Defenders' mirror

Zone transfer open → fix immediately. SPF/DMARC configured → spammers blocked. Internal hostnames hidden → architecture not leaked. DNS hygiene is recon hygiene.

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[Google Dorking, Shodan & Censy...](#)